

Glasser's Master Theorem

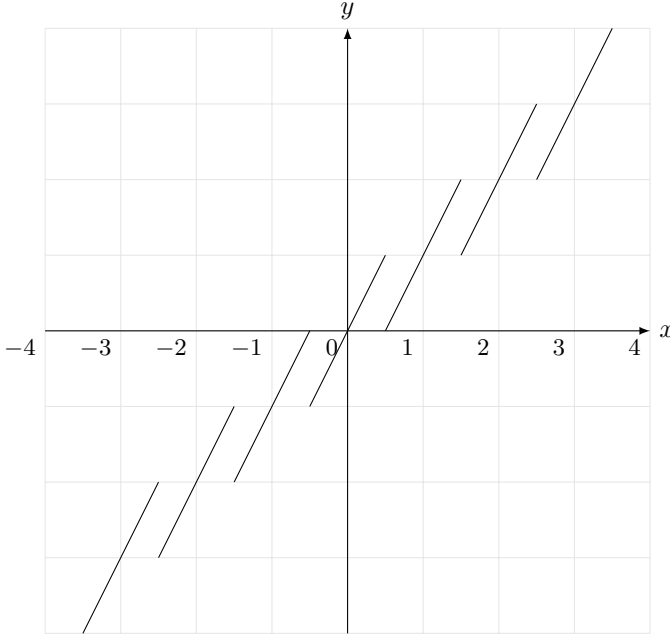
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Definition -1.1. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ **preserves the Lebesgue measure on $\mathbb{R}$** if for every Lebesgue measurable $A \subseteq \mathbb{R}$, we have $\mu(f^{-1}(A)) = \mu(A)$, where $f^{-1}(A) = \{x \in \mathbb{R} \mid f(x) \in A\}$ is the pre-image of A under f .

Examples.

1. For any $a \in \mathbb{R}$, the function $f(x) = x + a$ preserves the Lebesgue measure on $\mathbb{R}$.
2. The function $f(x) = 2x + [x] - [2x]$ preserves the Lebesgue measure on $\mathbb{R}$. This is because for each $n \in \mathbb{Z}$, the interval $[n, n+1)$ is mapped to two copies of itself, each with a slope of 2 (so the pre-image of any subinterval of $[n, n+1)$ is mapped to two subintervals, each with half the length). Likewise, for any $m \in \mathbb{Z} \setminus \{0\}$, the function $f(x) = mx + [x] - [mx]$ preserves the Lebesgue measure on $\mathbb{R}$.



0 The Cauchy-Schlömilch Transformation

Lemma 0.1 (Cauchy-Schlömilch Transformation). Suppose $c > 0$ and define $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ as follows:

$$f(x) = x - \frac{c}{x}$$

Then f preserves the Lebesgue measure on $\mathbb{R}$.

Proof. Suppose $t \in \mathbb{R}$. We first solve the inequality $f(x) < t$, i.e. $x - \frac{c}{x} < t$.

- If $x > 0$, this implies $x^2 - tx - c < 0$, and so $\frac{t - \sqrt{t^2 + 4c}}{2} < x < \frac{t + \sqrt{t^2 + 4c}}{2}$. Since $\frac{t - \sqrt{t^2 + 4c}}{2} < 0$, this simplifies to $0 < x < \frac{t + \sqrt{t^2 + 4c}}{2}$.
- If $x < 0$, this implies $x^2 - tx - c > 0$, and so $x < \frac{t - \sqrt{t^2 + 4c}}{2}$ or $x > \frac{t + \sqrt{t^2 + 4c}}{2}$. Since $\frac{t + \sqrt{t^2 + 4c}}{2} > 0$, this simplifies to $x < \frac{t - \sqrt{t^2 + 4c}}{2}$.

All in all, we have:

$$f(x) < t \iff x < \frac{t - \sqrt{t^2 + 4c}}{2} \text{ or } 0 < x < \frac{t + \sqrt{t^2 + 4c}}{2}$$

Similarly:

$$f(x) > t \iff \frac{t - \sqrt{t^2 + 4c}}{2} < x < 0 \text{ or } x > \frac{t + \sqrt{t^2 + 4c}}{2}$$

Note that for all $t \in \mathbb{R}$, we have:

$$\frac{\partial}{\partial t} \left(\frac{t \pm \sqrt{t^2 + 4c}}{2} \right) = \frac{1}{2} \pm \frac{t}{2\sqrt{t^2 + 4c}} \geq \frac{1}{2} - \frac{|t|}{2\sqrt{t^2 + 4c}} > \frac{1}{2} - \frac{1}{2} = 0$$

Thus the functions $t \mapsto \frac{t \pm \sqrt{t^2 + 4c}}{2}$ are both strictly increasing in t .

Now suppose $\alpha, \beta \in \mathbb{R}$, $\alpha < \beta$. Then we have:

$$\alpha < f(x) < \beta \iff \left(x < \frac{\beta - \sqrt{\beta^2 + 4c}}{2} \text{ or } 0 < x < \frac{\beta + \sqrt{\beta^2 + 4c}}{2} \right)$$

$$\text{and } \left(\frac{\alpha - \sqrt{\alpha^2 + 4c}}{2} < x < 0 \text{ or } x > \frac{\alpha + \sqrt{\alpha^2 + 4c}}{2} \right)$$

This reduces to:

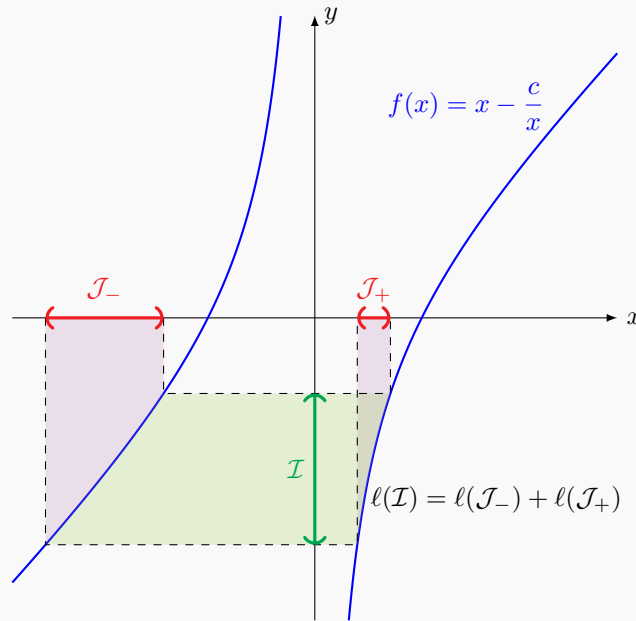
$$\frac{\alpha - \sqrt{\alpha^2 + 4c}}{2} < x < \frac{\beta - \sqrt{\beta^2 + 4c}}{2} \text{ or } \frac{\alpha + \sqrt{\alpha^2 + 4c}}{2} < x < \frac{\beta + \sqrt{\beta^2 + 4c}}{2}$$

The union of these intervals has length:

$$\frac{\beta - \sqrt{\beta^2 + 4c}}{2} - \frac{\alpha - \sqrt{\alpha^2 + 4c}}{2} + \frac{\beta + \sqrt{\beta^2 + 4c}}{2} - \frac{\alpha + \sqrt{\alpha^2 + 4c}}{2} = \frac{\beta}{2} - \frac{\alpha}{2} + \frac{\beta}{2} - \frac{\alpha}{2} = \beta - \alpha$$

Thus $\mu(f^{-1}((\alpha, \beta))) = \mu((\alpha, \beta))$ for all $\alpha, \beta \in \mathbb{R}$, $\alpha < \beta$. By the [Carathéodory extension theorem](#), we have $\mu(f^{-1}(A)) = \mu(A)$ for every Lebesgue measurable $A \subseteq \mathbb{R}$. Thus f preserves the Lebesgue measure on $\mathbb{R}$. 🧐

Cauchy-Schlömilch Transformation



For any interval on the y -axis, the length of the interval is equal to the sum of the lengths of the corresponding intervals on the x -axis.

1 Glasser's Master Theorem

The following theorem was first stated in 1983 by [M. Lawrence Glasser](#) (1933–).

Theorem 1.1 (Glasser's Master Theorem). Suppose $a_1, a_2, \dots, a_n \in \mathbb{R}$, $c_1, c_2, \dots, c_n > 0$ and $b \in \mathbb{R}$. Define $f : \mathbb{R} \setminus \{a_1, a_2, \dots, a_n\}$ as follows:

$$f(x) = x - b - \frac{c_1}{x - a_1} - \frac{c_2}{x - a_2} - \dots - \frac{c_n}{x - a_n}$$

Then f preserves the Lebesgue measure on $\mathbb{R}$.

Proof. Assume without loss of generality that $a_1 < a_2 < \dots < a_n$. Define $a_0 = -\infty$ and $a_{n+1} = \infty$, and for each $k \in \{0, 1, \dots, n\}$, define $I_k = (a_k, a_{k+1})$. Note that for all $x \in \mathbb{R} \setminus \{a_1, a_2, \dots, a_n\}$, we have:

$$f'(x) = 1 + \frac{c_1}{(x - a_1)^2} + \frac{c_2}{(x - a_2)^2} + \dots + \frac{c_n}{(x - a_n)^2} > 1$$

We also have $\lim_{x \rightarrow a_k^-} f(x) = \infty$ and $\lim_{x \rightarrow a_k^+} f(x) = -\infty$ for all $k \in \{0, 1, \dots, n\}$. Thus for each $k \in \{0, 1, \dots, n\}$, the restriction of f to I_k is a bijection from I_k to $\mathbb{R}$.

Define $g_k : \mathbb{R} \rightarrow I_k$ as the inverse of the restriction of f to I_k . Then for each $y \in \mathbb{R}$, the equation $f(x) = y$ has exactly $n+1$ solutions, namely $g_0(y), g_1(y), \dots, g_n(y)$. Multiplying both sides of $f(x) - y = 0$ by $(x - a_1)(x - a_2) \cdots (x - a_n)$, we get:

$$(x - a_1)(x - a_2) \cdots (x - a_n) \left(x - b - \frac{c_1}{x - a_1} - \frac{c_2}{x - a_2} - \cdots - \frac{c_n}{x - a_n} \right) = 0$$

$$(x - a_1)(x - a_2) \cdots (x - a_n)(x - b - y) + P_{n-1}(x) = 0$$

Where $P_{n-1}(x)$ is a polynomial in x of degree at most $n-1$. Since the left side is a monic polynomial of degree $n+1$ that vanishes at $g_0(y), g_1(y), \dots, g_n(y)$, it must be equal to $(x - g_0(y))(x - g_1(y)) \cdots (x - g_n(y))$. Thus we have:

$$(x - a_1)(x - a_2) \cdots (x - a_n)(x - b - y) + P_{n-1}(x) = (x - g_0(y))(x - g_1(y)) \cdots (x - g_n(y))$$

Comparing the coefficients of x^n on each side, we get:

$$-(a_1 + a_2 + \cdots + a_n + b + y) = -(g_0(y) + g_1(y) + \cdots + g_n(y))$$

Since each g_k is strictly increasing, for any $y, z \in \mathbb{R}$, $y < z$, we have $f(x) \in (y, z)$ if and only if $x \in (g_k(y), g_k(z))$ for some $k \in \{0, 1, \dots, n\}$. This yields:

$$\begin{aligned} \mu(f^{-1}((y, z))) &= \mu\left(\bigcup_{k=0}^n (g_k(y), g_k(z))\right) \\ &= \sum_{k=0}^n \mu((g_k(y), g_k(z))) \\ &= \sum_{k=0}^n (g_k(z) - g_k(y)) \\ &= (a_1 + a_2 + \cdots + a_n + b + z) - (a_1 + a_2 + \cdots + a_n + b + y) \\ &= z - y \\ &= \mu((y, z)) \end{aligned}$$

Thus $\mu(f^{-1}((y, z))) = \mu((y, z))$ for all $y, z \in \mathbb{R}$, $y < z$. By the [Carathéodory extension theorem](#), we have $\mu(f^{-1}(A)) = \mu(A)$ for every Lebesgue measurable $A \subseteq \mathbb{R}$. Thus f preserves the Lebesgue measure on $\mathbb{R}$. 🧐

Remark. If $n = 1$, i.e. $f(x) = x - b - \frac{c}{x-a}$ (where $c > 0$), we can explicitly compute the inverse of f :

$$f^{-1}(y) = \frac{y + b + a \pm \sqrt{(y + b - a)^2 + 4c}}{2}$$

The $\pm$ corresponds to the fact that there are two values of x for each value of y , one with $x > b$ and one with $x < b$.

Theorem 1.2. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ preserves the Lebesgue measure on $\mathbb{R}$. Then for any $\phi \in L^1(\mathbb{R})$, we have:

$$\int_{\mathbb{R}} \phi(f(x)) d\mu(x) = \int_{\mathbb{R}} \phi(x) d\mu(x)$$

Proof. If $\phi = 1_A$ is an indicator function of some Lebesgue measurable $A \subseteq \mathbb{R}$, this reduces to $\mu(f^{-1}(A)) = \mu(A)$, which is true as f preserves the Lebesgue measure on $\mathbb{R}$ by assumption. By linearity, the result holds for simple functions (linear combinations of indicator functions), and by approximation, it holds for all integrable functions. 🧐

$$\int_{\mathbb{R}} \phi(f(x)) d\mu(x)$$

In the case of $f(x) = x - \frac{c}{x}$ (the Cauchy-Schlömilch transformation), we can give a much simpler proof:

Proof. Substituting $u = -\frac{c}{x}$, we get:

$$\int_{-\infty}^{\infty} \phi\left(x - \frac{c}{x}\right) dx = c \int_{-\infty}^{\infty} \frac{1}{u^2} \phi\left(u - \frac{c}{u}\right) du$$

We now add the left side of the above equation to both sides:

$$\begin{aligned} 2 \int_{-\infty}^{\infty} \phi\left(x - \frac{c}{x}\right) dx &= \int_{-\infty}^{\infty} \phi\left(x - \frac{c}{x}\right) dx + c \int_{-\infty}^{\infty} \frac{1}{x^2} \phi\left(x - \frac{c}{x}\right) dx \\ &= \int_{-\infty}^{\infty} \left(1 + \frac{c}{x^2}\right) \phi\left(x - \frac{c}{x}\right) dx \end{aligned}$$

We now substitute $y = x - \frac{c}{x}$. Note that as x runs from $-\infty$ to ∞ , y runs from $-\infty$ to ∞ *twice* (once when $x < 0$ and once when $x > 0$). This introduces a factor of 2 below:

$$2 \int_{-\infty}^{\infty} \phi\left(x - \frac{c}{x}\right) dx = 2 \int_{-\infty}^{\infty} \phi(y) dy$$

Dividing by 2 yields the desired result. 🤖

Example.

$$\int_{-\infty}^{\infty} \frac{x^2}{x^4 + 1} dx = \int_{-\infty}^{\infty} \frac{1}{\left(x - \frac{1}{x}\right)^2 + 2} dx = \int_{-\infty}^{\infty} \frac{1}{x^2 + 2} dx = \frac{\pi}{\sqrt{2}}$$

Example. Unfortunately, $\frac{1}{x^4+1}$ does not have a simple expression in terms of $x - \frac{1}{x}$. However, note that:

$$\frac{x^2 + 1}{x^4 + 1} = \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 2}$$

This allows us to easily find $\int_{-\infty}^{\infty} \frac{x^2+1}{x^4+1} dx$. Note that we are *not* using Glasser's master theorem here, we are simply substituting $u = x - \frac{1}{x}$.

$$\int_{-\infty}^{\infty} \frac{x^2 + 1}{x^4 + 1} dx = \int_{-\infty}^{\infty} \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 2} dx = 2 \int_{-\infty}^{\infty} \frac{1}{u^2 + 2} du = 2 \left(\frac{\pi}{\sqrt{2}} \right) = \sqrt{2}\pi$$

The factor of 2 comes from the fact that as x runs from $-\infty$ to ∞ , u runs from $-\infty$ to ∞ twice. Thus, we have:

$$\int_{-\infty}^{\infty} \frac{1}{x^4 + 1} dx = \int_{-\infty}^{\infty} \frac{x^2 + 1}{x^4 + 1} dx - \int_{-\infty}^{\infty} \frac{x^2}{x^4 + 1} dx = \sqrt{2}\pi - \frac{\pi}{\sqrt{2}} = \frac{\pi}{\sqrt{2}}$$

Example.

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{x^4}{x^8 + 1} dx &= \int_{-\infty}^{\infty} \frac{1}{\left(x - \frac{1}{x}\right)^4 + 4\left(x - \frac{1}{x}\right)^2 + 2} dx = \int_{-\infty}^{\infty} \frac{1}{x^4 + 4x^2 + 2} dx = \frac{\sqrt{4 - 2\sqrt{2}}}{4} \pi \\ \int_{-\infty}^{\infty} \frac{x^4}{x^8 + x^4 + 1} dx &= \int_{-\infty}^{\infty} \frac{1}{\left(x - \frac{1}{x}\right)^4 + 4\left(x - \frac{1}{x}\right)^2 + 3} dx = \int_{-\infty}^{\infty} \frac{1}{x^4 + 4x^2 + 3} dx = \frac{3 - \sqrt{3}}{6} \pi \\ \int_{-\infty}^{\infty} \frac{x^4}{x^8 + ax^4 + 1} dx &= \int_{-\infty}^{\infty} \frac{1}{\left(x - \frac{1}{x}\right)^4 + 4\left(x - \frac{1}{x}\right)^2 + a + 2} dx = \int_{-\infty}^{\infty} \frac{1}{x^4 + 4x^2 + a + 2} dx = \sqrt{\frac{\sqrt{a+2} - 2}{2(a^2 - 4)}} \pi \\ \int_{-\infty}^{\infty} \frac{x^6 + x^4}{x^{10} + 1} dx &= \int_{-\infty}^{\infty} \frac{1}{\left(x - \frac{1}{x}\right)^4 + 3\left(x - \frac{1}{x}\right)^2 + 1} dx = \int_{-\infty}^{\infty} \frac{1}{x^4 + 3x^2 + 1} dx = \frac{\pi}{\sqrt{5}} \\ \int_{-\infty}^{\infty} \frac{x^4}{(x^4 + 1)(x^4 - x^2 + 1)} dx &= \int_{-\infty}^{\infty} \frac{1}{\left(x - \frac{1}{x}\right)^4 + 3\left(x - \frac{1}{x}\right)^2 + 2} dx = \int_{-\infty}^{\infty} \frac{1}{x^4 + 3x^2 + 2} dx = \left(1 - \frac{1}{\sqrt{2}}\right) \pi \end{aligned}$$

Example.

$$\int_{-\infty}^{\infty} e^{-(x - \frac{1}{x})^2} dx = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

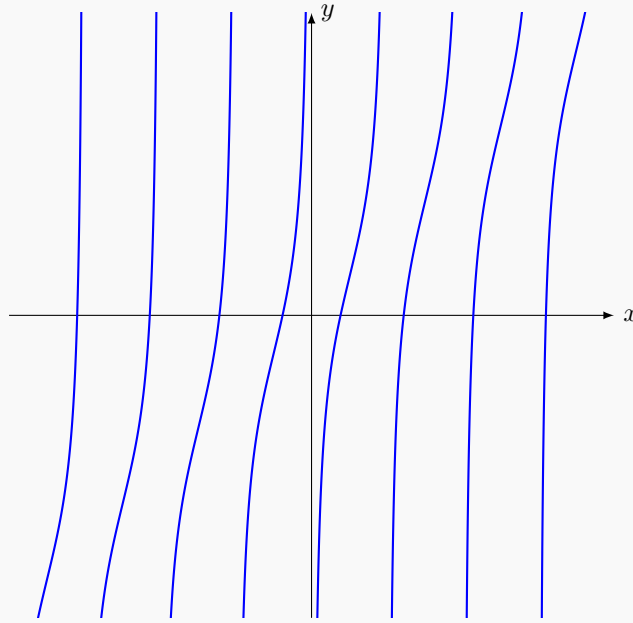
Example.

$$\int_{-\infty}^{\infty} \operatorname{sech}\left(x - \frac{1}{x}\right) dx = \int_{-\infty}^{\infty} \operatorname{sech}(x) dx = \pi$$

2 Generalization to infinitely many poles

We can generalize Glasser's master theorem to more exotic functions.

The function $f(x) = x - \cot(x)$



The function $f : \mathbb{R} \setminus \pi\mathbb{Z} \rightarrow \mathbb{R}$, $f(x) = x - \cot(x)$ is qualitatively very similar to the functions in Glasser's master theorem (except that this function has *infinitely* many poles).

It turns out that f also preserves the Lebesgue measure on $\mathbb{R}$. This is because:

$$\cot(x) = \cdots + \frac{1}{x-3\pi} + \frac{1}{x-2\pi} + \frac{1}{x-\pi} + \frac{1}{x} + \frac{1}{x+\pi} + \frac{1}{x+2\pi} + \frac{1}{x+3\pi} + \cdots$$

More formally, for all $x \in \mathbb{R} \setminus \pi\mathbb{Z}$, we have:

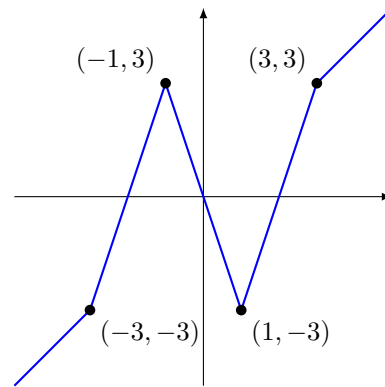
$$\cot(x) = \lim_{n \rightarrow \infty} \sum_{k=-n}^n \frac{1}{x+k\pi}$$

<https://math.stackexchange.com/a/1015739>

There are many more functions that preserve the Lebesgue measure on $\mathbb{R}$. The example below is continuous and piecewise linear.

Example. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ as follows:

$$f(x) = \begin{cases} x & x < -3 \text{ or } x > 3 \\ 3x+6 & -3 < x < -1 \\ -3x & -1 \leq x \leq 1 \\ 3x-6 & 1 < x < 3 \end{cases}$$



Then f is continuous, piecewise linear and preserves the Lebesgue measure on $\mathbb{R}$. Informally, this is because $f(x) = x$ except on $(-3, 3)$, where it traverses the interval $(-3, 3)$ three times, each with a slope of ± 3 . More examples can be constructed similarly, e.g. by traversing an interval 5 times, each with a slope of ± 5 .